

ASTRALINK — AI-BASED VIRTUAL CAMERA TRACKING SYSTEM FOR MOBILE FS

Coarse Alignment - Point-in-Time Performance Report

T = 1.52 s

1. TEST CONFIGURATION

Scenario: synthetic_fs0 Target: TGT-01 Camera: 640 x 480 FOV: 4.0 deg x 3.0 deg Update Rate: 30.0 Hz

Noise: None 10.00 Camera jitter: OFF 5.00 px/frame Platform motion: OFF 10.00 px Turbulence: OFF 10.00 px

2. PERFORMANCE

Processing FPS: 296.7 (>= 20 FPS) Average Tracking Error: 2.68 px Maximum Tracking Error: 6.08 px (<= 10 px)

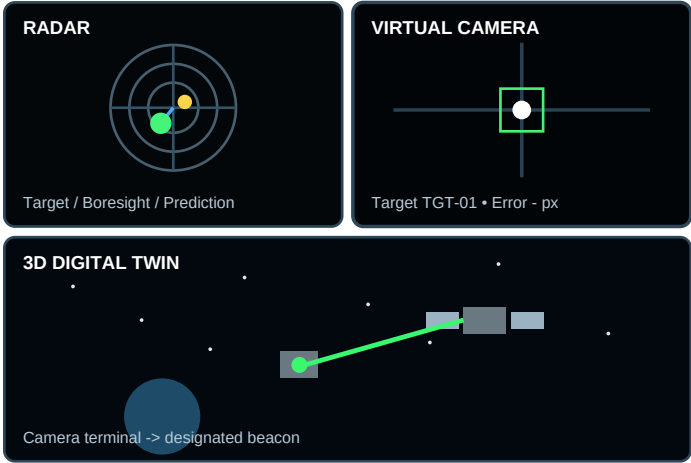
Acquisition Time: 0.03 s (<= 2 s) Re-acquisition: - s (<= 1 s) Lock Retention: 100.00 %

RMSE: 2.97 px Target Loss: 0.00 % Avg Processing: 1.73 ms Re-acquisitions: 0

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Update Rate	30.0 Hz	>= 20 Hz	PASS
Processing Speed	296.7 FPS	>= 20 FPS	PASS
Acquisition Time	0.03 s	<= 2 s	PASS
Tracking Error (Max)	6.08 px	<= 10 px	PASS
Target Loss	0.00 %	< 5 %	PASS
Re-acquisition	- s	<= 1 s	-

4. VISUAL SNAPSHOT (T = 1.52 s)



State: BENCHMARK Pan: -1.17 deg Tilt: -1.17 deg AI: -

5. SYSTEM SUMMARY

COARSE ALIGNMENT CONTINUES - HANDOFF NOT READY

Designated target TGT-01 at 1.52 s. Detection, AI verification, Kalman prediction and pan/tilt control are reported from the recorded telemetry snapshot.